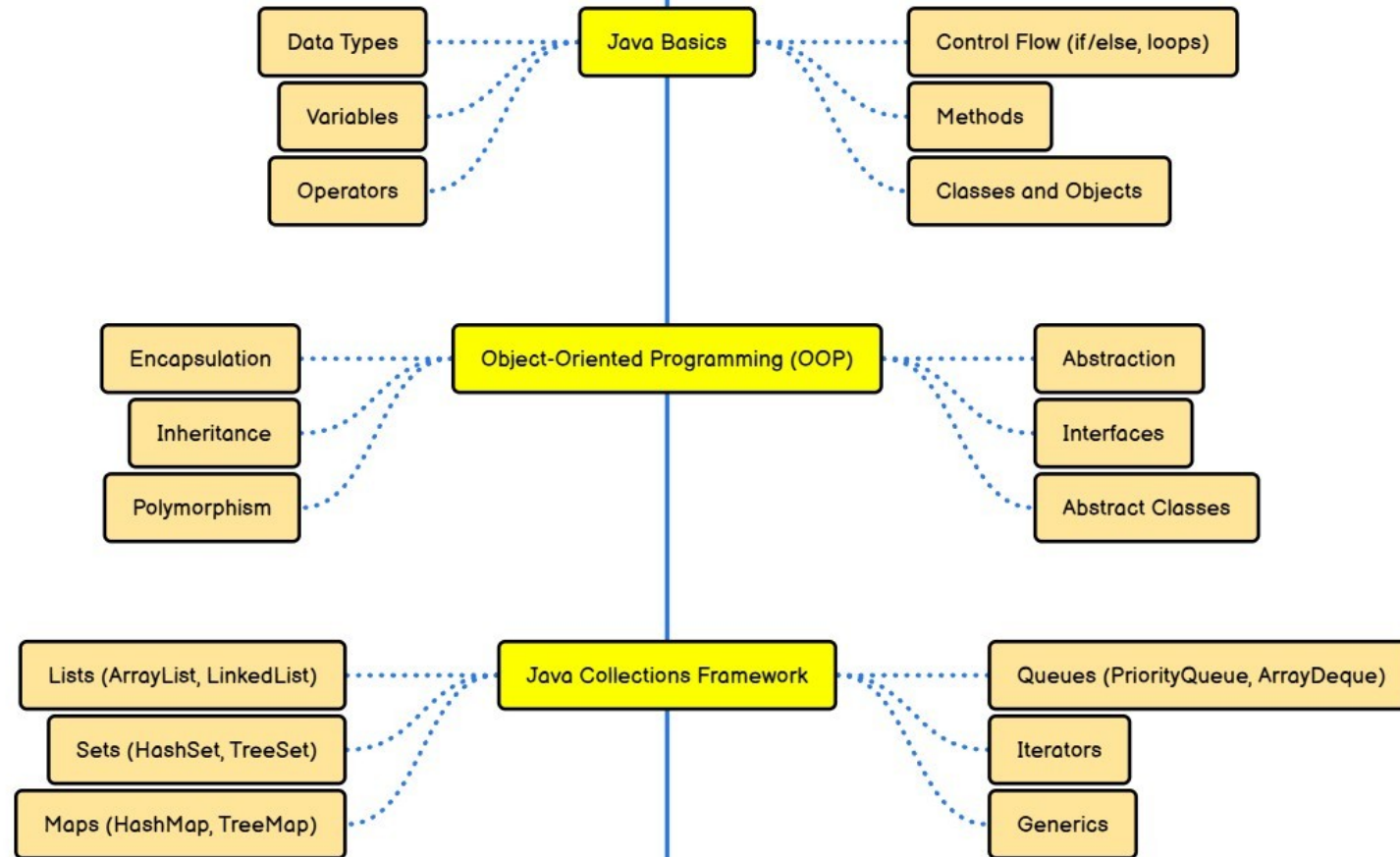


DSA with Java

Foundations of Java



Algorithm Analysis

Java Memory Management

Stack vs Heap

Garbage Collection

Memory Leaks

Object Lifecycles

JVM Internals

Profiling Tools

Asymptotic Notation

Big O Notation

Big Omega Notation

Big Theta Notation

Little o Notation

Little omega Notation

Time Complexity Analysis

Space Complexity Analysis

Auxiliary Space

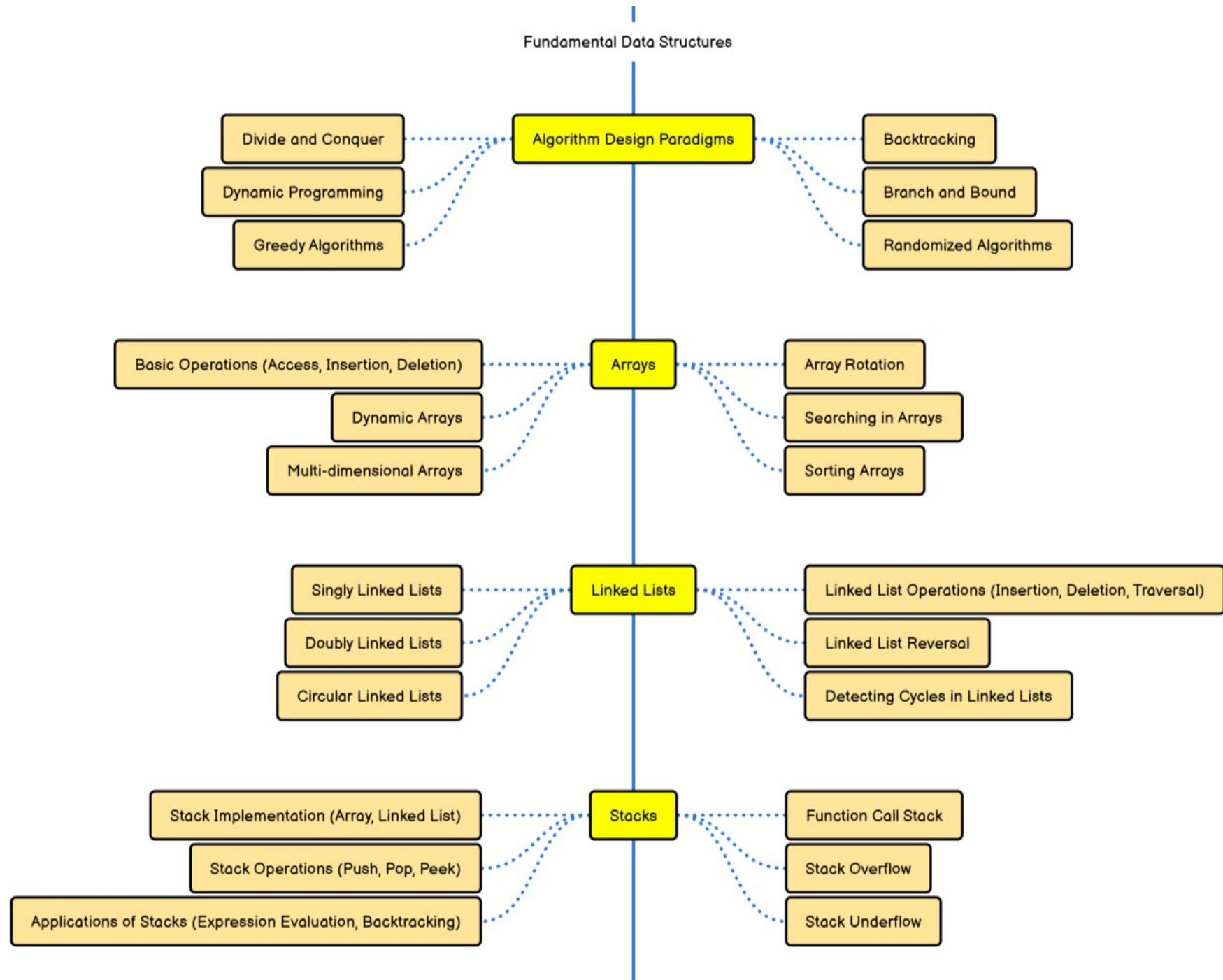
Space Complexity of Data Structures

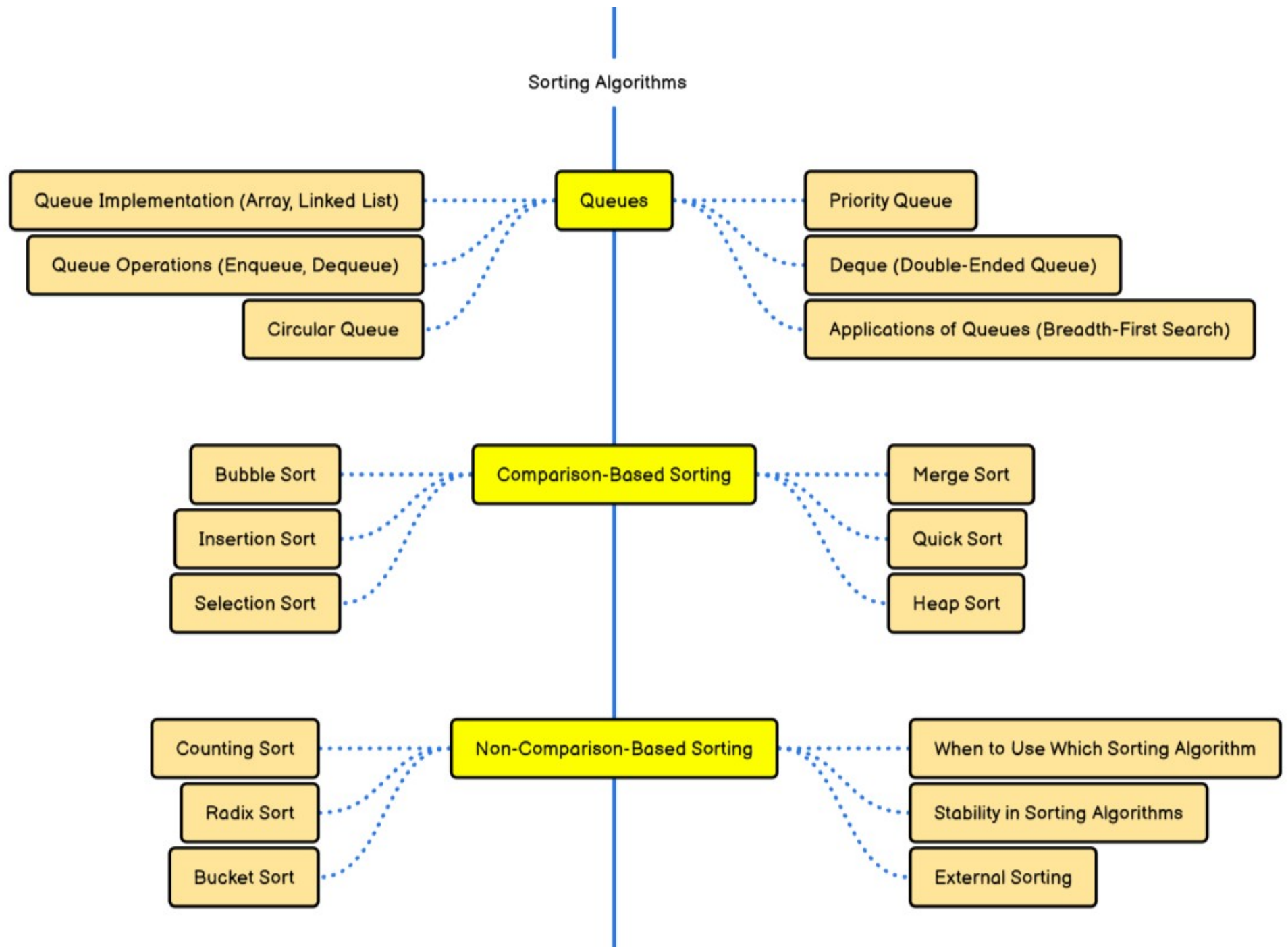
Space-Time Tradeoff

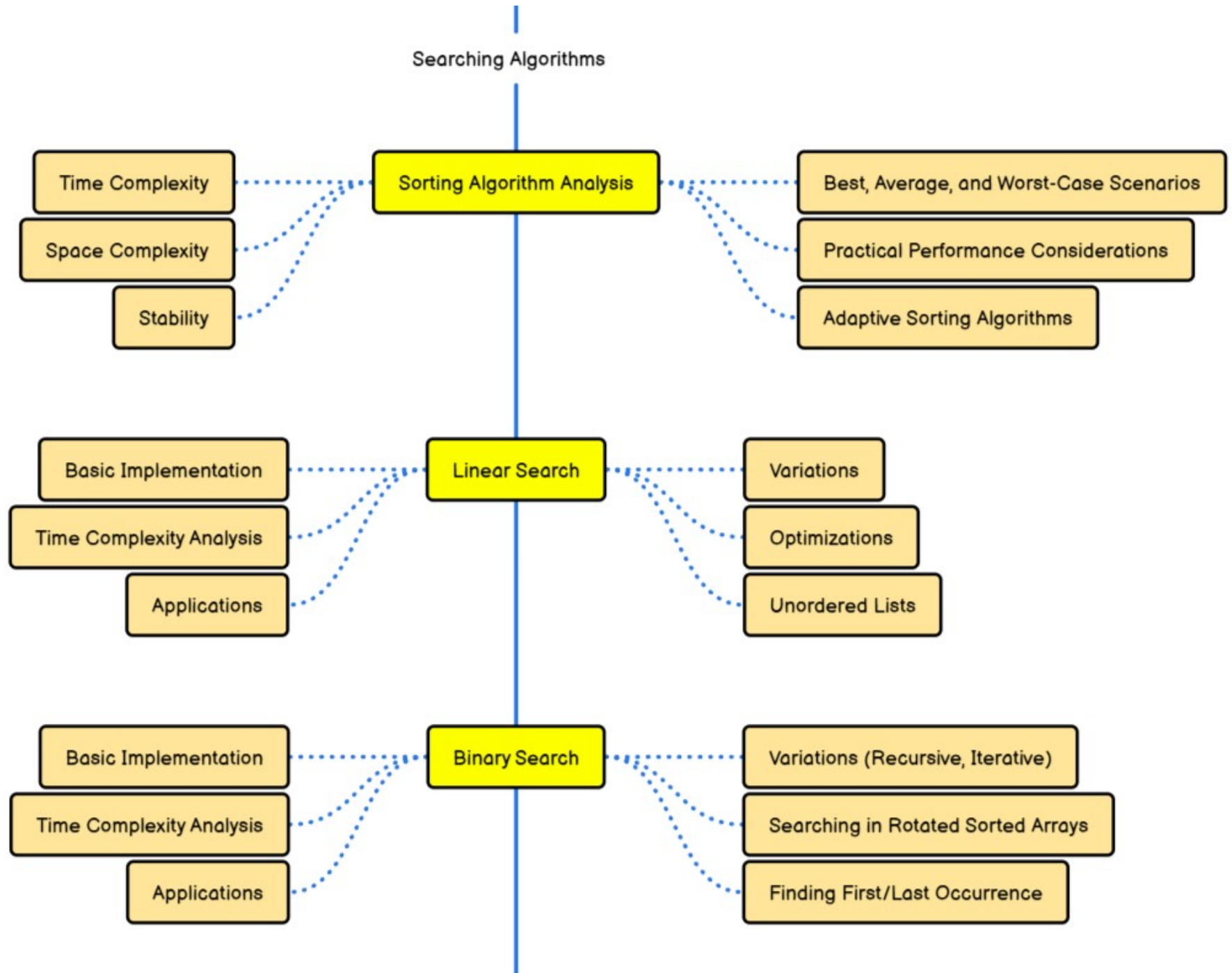
Best, Average, and Worst-Case Analysis

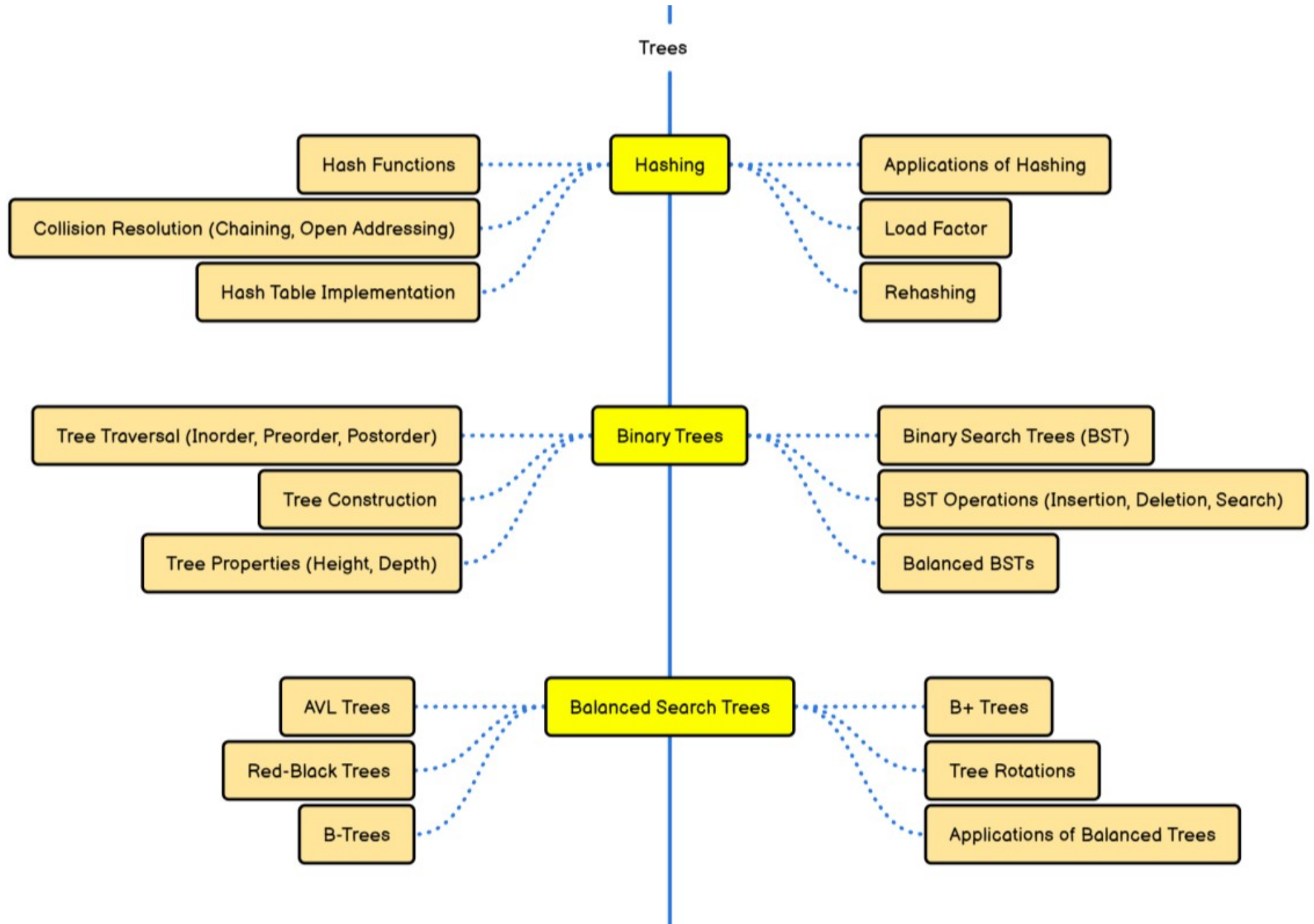
Amortized Analysis

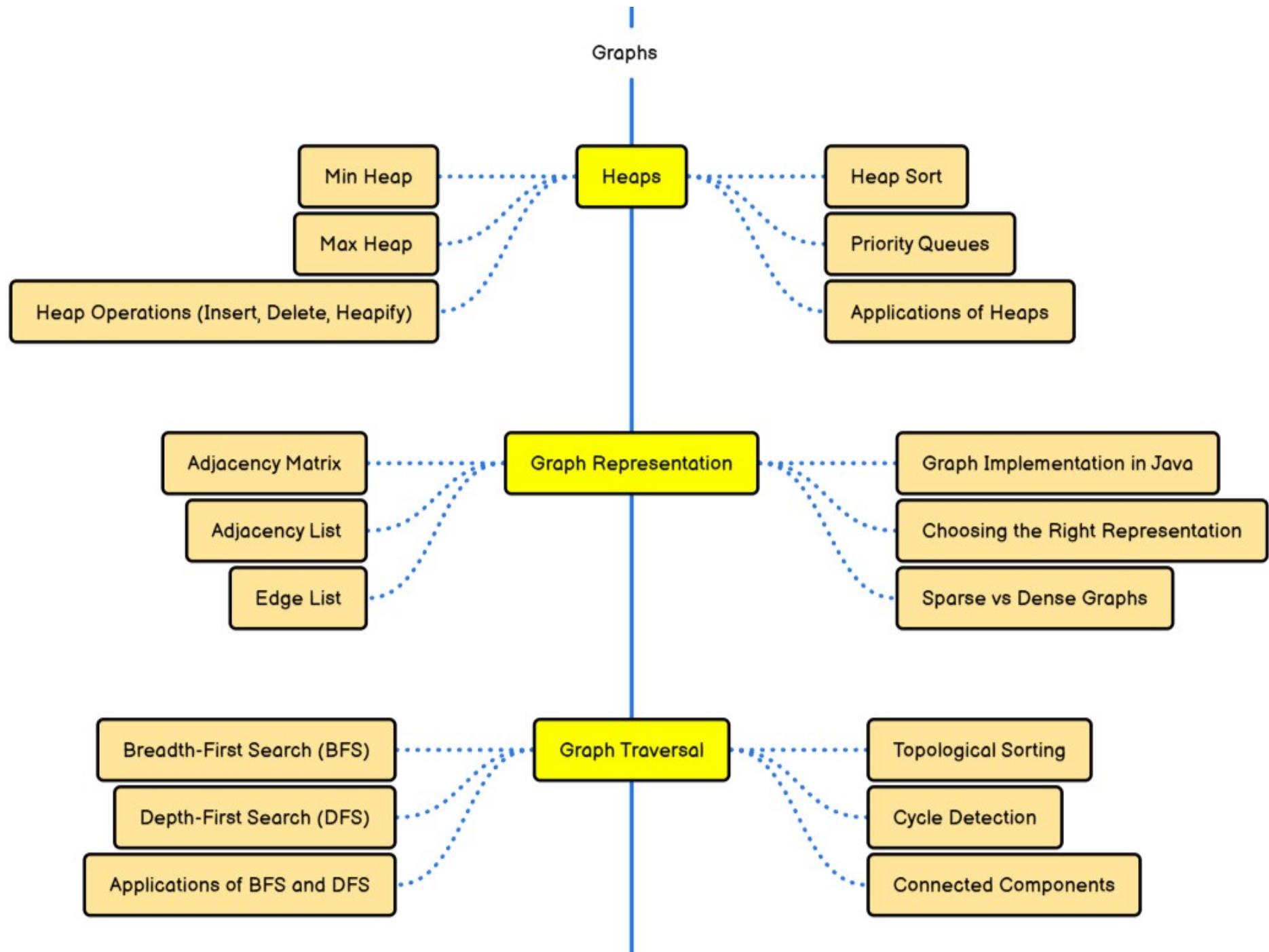
Practical Considerations

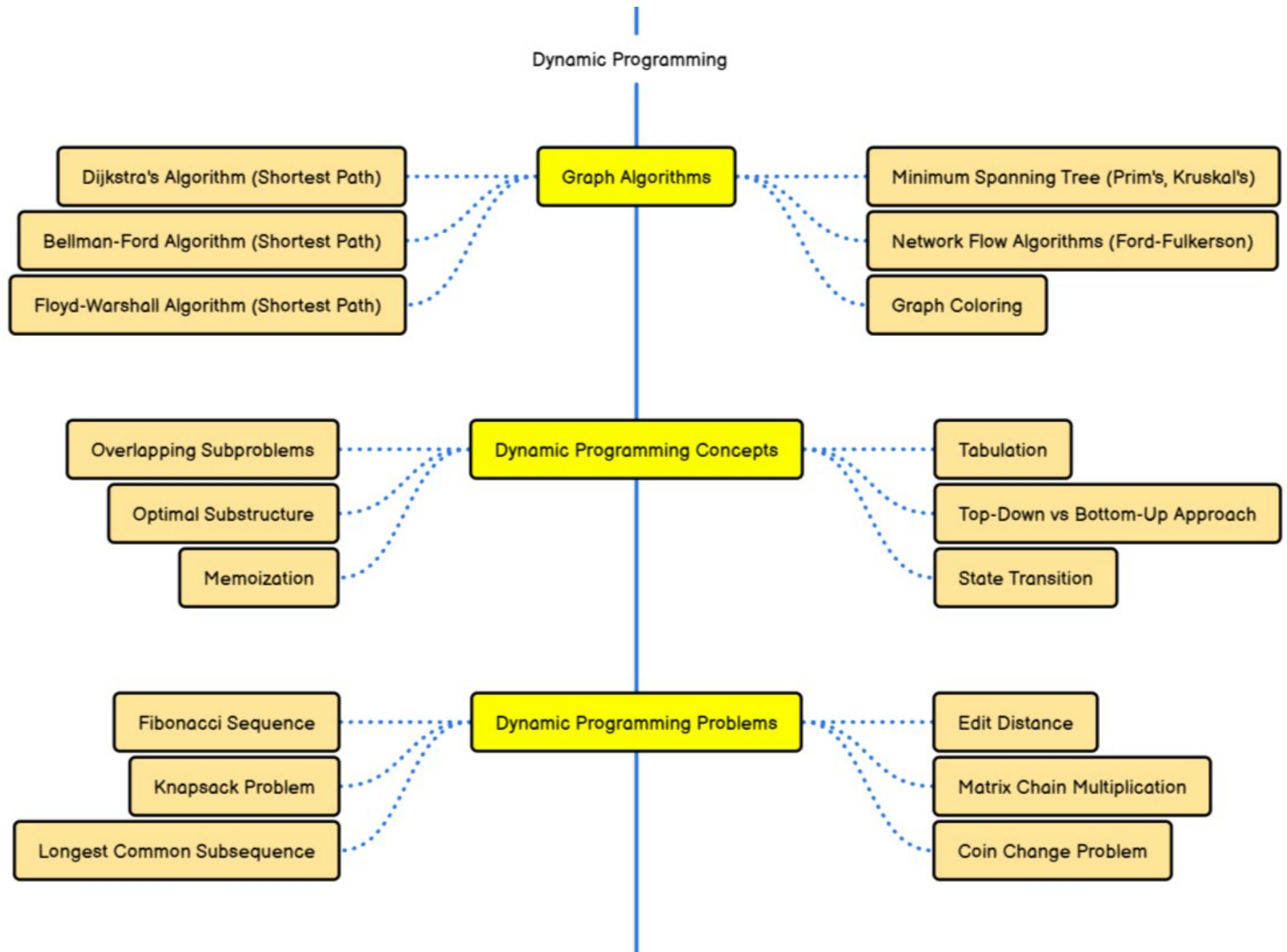












Greedy Algorithms

Advanced Dynamic Programming

Bitmasking DP

DP on Trees

DP with State Compression

Optimizations in DP

Divide and Conquer DP

Applications of DP

Greedy Algorithm Concepts

Optimal Choice

Feasible Solution

Greedy Choice Property

Optimal Substructure

When to Use Greedy Algorithms

Limitations of Greedy Algorithms

Greedy Algorithm Problems

Activity Selection Problem

Fractional Knapsack Problem

Huffman Coding

Job Sequencing Problem

Minimum Spanning Tree (Prim's, Kruskal's)

Dijkstra's Algorithm

Backtracking

